# Mastara (ማስታራ) — Software Requirements Specification v2.3

*Merged rebuild baseline · Functional requirements by partition · UI flow design · Commerce, device, offline and admin operations*

| Field | Value |
|---|---|
| Product | Mastara (ማስታራ) — bilingual, offline-first learning app for Ethiopian university freshmen |
| Document ID | MASTARA-SRS-001 |
| Version | 2.3 — Merged Rebuild (supersedes v1.2, v1.3, EthioLearn v1.0, Freshman+ v1.0) |
| Status | Draft for product-owner approval |
| Date | 06 October 2026 |
| Repository | Kewakibt-Enterprise-Platform/Mastara |
| Student client | React Native with Expo (Android first, iOS supported) |
| Admin platform | Web portal built with Next.js |
| Backend | NestJS |
| Database | PostgreSQL 16 or newer — replaces MySQL used in v2.0 to v2.2. Package manager: npm (workspaces) |
| Roles | STUDENT, ADMIN (Admin portal contains every operational asset) |
| Currency / Languages | ETB · English + Amharic |
| App package target | Under 50 MB (hard release gate) |
| Normative language | MUST = mandatory · SHOULD = recommended · MAY = optional |

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# 1. Purpose, Source Reconciliation and Change Baseline

## 1.1 Purpose

This document is the single authoritative specification for Mastara. It merges the four earlier documents and every instruction given in the product brief, then organises functional requirements into **partitions** (Identity, Catalog, Content, Assessment, Commerce, Offline, Admin), each with a workflow, UI flow and acceptance criteria.

## 1.2 What each source contributed

| Source | Kept in v2.0 | Not carried forward |
|---|---|---|
| Mastara SRS v1.2 | Student and Admin UI flows, device binding, receipt lifecycle, exam integrity model, sync protocol, conflict matrix, QA gates, FR-001 to FR-066 numbering | 20 MB app target |
| Mastara SRS v1.3 | Front-end workflow, Home and Course tab structure, promotional unpaid courses, automatic expiry, scan-to-PDF, error codes, lifecycles, roadmap | Redis as mandatory |
| EthioLearn v1.0 | Architecture depth: sync protocol, DRM and content protection, observability, DR and backup, risk register, decision log | GPA, study plans, AI tutor, usage-based billing, recommendation engine (all deferred or out of scope) |
| Freshman+ v1.0 | Normative language, two-role authorization formula, admin boundary rule, page-flow discipline, success indicators | GPA calculator |

| Source | Kept in v2.0 | Not carried forward |
|---|---|---|
| Product brief (this session) | PostgreSQL, Expo, Next.js, course bundles and pricing, stream filter, 6+ image to PDF, standalone cross-course COC Exam, free-trial portions, device-integrated app | — |

## 1.3 Authoritative decisions

| # | Decision | Rule |
|---|---|---|
| D1 | Technology | React Native (Expo) mobile, Next.js admin web, NestJS API, PostgreSQL 16+ |
| D2 | App size | Installed package MUST be under 50 MB; replaces the earlier 20 MB target. Runtime memory stays under 120 MB on 2–3 GB RAM phones |
| D3 | Roles | Only STUDENT and ADMIN. Admin is the only privileged role and owns the assets of any other role (content editor, reviewer, finance, support) |
| D4 | Device | 1 Student account = 1 bound physical device. The app MUST NOT work on another device while bound |
| D5 | GPA | Out of scope (removed everywhere) |
| D6 | Payment | Manual receipt screenshot via CBE Mobile Banking, CBE Birr or Telebirr, verified by Admin |
| D7 | Access | Entitlements per course or package, with Admin-managed duration and automatic server-side expiry |
| D8 | Amharic notes | One logical bilingual content model. No duplicate Amharic-only source files |

| # | Decision | Rule |
|---|---|---|
| D9 | COC independence | COC Exam is independent of Exams: a standalone module opened only from the Home COC Exam card, with one question pool across all courses. Exams keeps exactly Mid, Final and Mock, as in the approved mockups |
| D10 | Scan to PDF | Admin converts one or more scanned images (6 or more fully supported) into one PDF in Admin-chosen order |

**Added in v2.1 after the UI mockup review**

- **D11 Profile-first onboarding:** after registration or first login the student lands on the Profile page to set university, program, semester, stream and courses, then goes to Home. The separate Universities, Program and Semester screens are replaced by this page.
- **D12 My Course List:** the student adds and removes courses from Profile. The Course tab shows only that list. Adding a course does not grant access; only a purchase does.
- **D13 Screen design baseline:** Course, Material, Tests and Exams screens follow the approved mockups (section 5.10). Bottom navigation is Home, Course, Progress, Profile.
- **D14 COC independent:** COC Exam is a standalone Home module with a question pool across all courses. It is not a tab of Exams and not tied to one course.
- **D15 Free trial:** unpaid students can add courses to My Course List and use an Admin-flagged portion of each course (section 3.8).

**Added in v2.3**

- **D16 Database:** PostgreSQL 16 or newer replaces MySQL. Use `uuid` keys, `timestamptz`, `jsonb` and native enums or checks; `UNIQUE(transaction_reference)` and per-course entitlement rows stay as specified.
- **D17 Package manager:** npm workspaces replace pnpm. Run Turborepo on top of npm.
- **D18 Card-based Home:** Home follows section 5.2.
- **D19 My Courses:** search, filter icon and the tabs All, Ongoing, Completed (section 5.11).
- **D20 Brand unchanged:** colors, fonts and logo stay as before.

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# 2. Product Definition, Scope and Roles

## 2.1 Definition

Mastara helps freshmen move from finding materials to understanding them, practising, and sitting exams, even on cheap phones with unreliable internet. Students choose University, Program and Semester, buy access to the courses they need, and learn from notes, Amharic notes, quizzes, exams, plus a standalone COC Exam module.

## 2.2 In scope

- Registration, login, secure session and one-device binding with student-initiated release
- University, Program, Semester selection and stream-filtered course catalog (Natural, Social, Common)
- Course tabs: Material, Tests, Exams; COC Exam as a standalone Home module; free trial for unpaid students
- Notes, Amharic notes, modules, PDFs, bilingual STEM rendering with KaTeX
- Practice and Quiz 1, 2, 3 with immediate feedback; Mid, Final and Mock exams; standalone cross-course COC Exam
- Bundles, single courses, extra courses, Full Freshman package, receipt payment and Admin verification
- Admin-defined durations and automatic expiry; promotional unpaid courses
- Offline-first storage, outbox sync, monotonic exam timer
- Admin web portal covering every operational function, including scan-to-PDF
- Notifications, support, audit and reports

## 2.3 Out of scope

GPA calculation; automated payment gateways; live remote proctoring; AI tutor and recommendation engine (deferred to a later release); registrar-grade student information system; roles other than Student and Admin; iOS-specific store billing.

### 2.4 Roles and authorization

| Role | Responsibility |
|---|---|
| STUDENT | Register, bind device, select academic scope, buy access, learn, practise, sit permitted exams, upload payment proof, release own device, receive notices |
| ADMIN | Everything else: students, catalog, content, scanning, questions, exams, COC, products, prices, durations, payments, entitlements, devices, notifications, support, audit, reports, settings |

Authorization rule (server-side, every request):

`ALLOW = authenticated + role permission + resource scope + resource state + business rule`

Example: `CAN_OPEN_CONTENT = STUDENT AND device_matches AND entitlement_active(course) AND content.status = PUBLISHED`.

The client never sends or selects a role. The backend resolves it from stored identity. A forged role field is ignored and logged.

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## 3. Commerce and Access Model (Product Rules)

This partition defines what students can buy and how access is derived. Every number is **Admin-managed configuration**; the values shown are defaults from the brief.

### 3.1 Course classification

Every course carries one **stream**: `NATURAL`, `SOCIAL` or `COMMON`. Common courses belong to both streams. The student catalog is **filtered by stream**; Common courses appear in both filters.

## 3.2 Products

| Product | Default price | Rule |
|---|---|---|
| Semester Bundle | 200 ETB | Student selects between **7 and 10** courses from the chosen semester and stream (min and max Admin-configurable). One flat price per semester |
| Extra Course | 100 ETB, then 50 ETB | After the bundle maximum, each additional course costs the Admin-defined extra price. A tiered schedule (first extra, later extras) is supported |
| Single Course | Admin-defined | A student may buy one course **before** buying a bundle |
| Full Freshman Package | 350 ETB per stream | All courses of the stream plus Common courses for the whole freshman year (both semesters), with COC and exams |
| Both Streams | Admin-defined (default 2 × 350) | Student may buy Natural, Social, or both streams at once |

## 3.3 What a paid course unlocks

A paid course unlocks **all** of its protected assets: modules, notes, Amharic notes, PDFs, Practice and Quizzes, Mid, Final and Mock exams, COC Exam access and university-specific material. Students who bought specific courses MUST NOT access unpaid courses. Unpaid courses stay visible as advertisement (see 3.5).

## 3.4 Duration and expiry

Each product has an Admin-managed `duration_days`. Approval creates entitlements with `starts_at` and `expires_at`. The server is authoritative: when `expires_at` passes, protected content locks automatically. Device clock changes and offline caches MUST NOT extend access.

## 3.5 Promotional unpaid courses

Admin may flag any course as promotional. Unpaid students see its title, description, stream and price, can **add it to My Course List**, and can use its Admin-set **free-trial portion** (section 3.8). Everything outside that portion stays locked with a Purchase action.

## 3.6 Purchase flow summary

`Store → choose Stream → choose product (Bundle / Single / Full) → select courses (Bundle: 7–10) → price summary → payment instructions → pay → upload receipt → reference → submit → Admin queue → Approve → entitlements active → Expiry → lock → Renew`

## 3.7 Pricing rules (testable)

| ID | Rule |
|---|---|
| PR-1 | Bundle submission with fewer than the minimum or more than the maximum selected courses is rejected client-side and server-side |
| PR-2 | Selecting courses beyond the maximum is only possible by adding Extra Courses; the price summary lists each extra price |
| PR-3 | Price is computed on the server from the live catalog and Admin configuration; the client price is display-only |
| PR-4 | A Full Freshman Package of a stream supersedes and covers any single or bundle purchases for the same stream; no double-charge prompt is shown for owned courses |
| PR-5 | Courses already actively entitled cannot be bought again until within the renewal window |
| PR-6 | Price changes apply to new orders only; pending orders keep their quoted price |
| PR-7 | Single-course purchases are not credited toward a later bundle unless Admin enables `credit_single_to_bundle` (default off) |

## 3.8 Free Trial for Unpaid Students

Unpaid students can add any course of their scope to My Course List and try a portion of it before paying. The portion is Admin-managed per course and enforced on the server.

| Area | Free-trial behaviour (default) |
|---|---|
| Modules, Notes, Amharic | Items flagged free open fully (default: the first chapter of each module); other items show a lock |

| Area | Free-trial behaviour (default) |
|---|---|
| Tests (Quiz 1, 2, 3) | Free-flagged chapters open with full feedback (default: the first chapter of each quiz); results are saved |
| Exams (Mid, Final, Mock) | Locked, unless Admin flags one sample paper per category as free |
| COC Exam (Home) | A sample set with an Admin-set number of questions; the full COC pool needs paid access (FR-184) |
| Duration | No time limit by default; Admin may set a trial expiry in days per course |
| After purchase | Trial progress and answers carry over to the paid course |

Trial items still require login and a bound device. A trial never creates an entitlement, so it cannot unlock anything outside the flagged portion. Trial items may be cached offline and are removed from the cache if Admin withdraws the flag.

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# 4. Information Architecture and Navigation

## 4.1 Hierarchy

`University → Program → Semester → Course (stream) → Material | Tests | Exams`

| Level | Structure |
|---|---|
| Home | Card-based: Header · Continue Learning · My Courses · Learning Stats · Store Status · COC Exam · Recently Accessed |
| Bottom navigation | Home · Course · Progress · Profile |
| Course (top tabs) | **Material** · **Tests** · **Exams** |
| Material (lower tabs) | **Modules** · **Notes** · **Amharic** (Amharic Note) |

| Level | Structure |
|---|---|
| Tests (lower tabs) | **Quiz 1** (Practice & Quiz 1) · **Quiz 2** · **Quiz 3**; each lists its chapters |
| Exams (lower tabs) | **Mid** · **Final** · **Mock**; each lists past exams by university |
| COC Exam (from Home only) | Standalone module, independent of Exams and of any one course; questions come from all courses |
| Profile | Personal info · University, Program, Semester, Stream · My Courses (add or remove) · Subscription and Store · Security and Device · Settings · Support |
| Free trial | Unpaid students use Admin-flagged items inside courses in My Course List; everything else is locked |

## 4.2 Student bottom navigation

Home · Course · Progress · Profile, as in the mockups. Progress is the Learning Status screen. Store, Notifications and Support open from Home cards, lock prompts and Profile.

## 4.3 Admin navigation

Dashboard · Students · Universities · Programs · Semesters · Courses · Content · Scan to PDF · Questions · Tests and Exams · COC · Products and Pricing · Payment Queue · Entitlements · Devices · Notifications and Support · Reports · Settings · Audit Logs

## 4.4 UI access states (all content cards)

| State | Presentation | Primary action |
|---|---|---|
| Available | Normal card | Open |
| Promotional | Ad badge, Free badge on trial items | Buy |
| Locked | Lock icon and reason | Buy or Renew |
| Pending | Verification indicator | View status |

| State | Presentation | Primary action |
|---|---|---|
| Expiring | Warning with days left | Renew |
| Expired | Locked, expiry date | Renew |
| Offline | Offline badge | Continue permitted work |

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# 5. Student UI Flow Design

## 5.1 Primary workflow

**Install → Splash → Login/Register → Device Registration → Profile Setup (university, program, semester, stream, courses) → Home**

| Step | System behaviour | On success | On failure |
|---|---|---|---|
| Install | Base package under 50 MB; no textbooks or PDFs bundled | Launch | — |
| Splash | Load config, bundled fonts, local DB, secure device ID, session | Login/Register, or Home if session, device and profile are valid | Offline splash shows cached state |
| Login/Register | Authenticate; register creates a STUDENT | Device Registration if unbound | Field errors; 429 lockout message |
| Device Registration | Read hardware UUID from Android Keystore or iOS Keychain; bind to account | Profile Setup if the profile is incomplete, else Home | 403 `DEVICE_MISMATCH_LOCKED` modal |

| Step | System behaviour | On success | On failure |
|---|---|---|---|
| Profile Setup | Mandatory on first login or right after registration: choose university, program, semester, stream, then courses (section 5.9) | Save → Home | Required-field errors; save retried offline |
| Home | Load learning, store and access state for the saved profile | Continue or browse | Cached Home with offline badge |

A returning student with a complete profile goes straight from Device check to Home. Profile Setup reappears only if a required field is empty.

## 5.2 Home layout

The Home screen is **card-based**. Cards stack in this order. Brand colors and typography are unchanged.

| Card | Content | Action |
|---|---|---|
| Header | Greeting "Hello, [first name]", line "Keep learning, your future is bright!", notification bell (dot when unread), profile avatar | Bell opens Notifications; avatar opens Profile |
| Continue Learning | Primary blue card: course, chapter, progress bar and percent | Resume last permitted topic, quiz or exam attempt (entitlement rechecked) |
| My Courses | Quick card: number of courses in My Course List | Opens the Course tab |
| Learning Stats | Quick card: chapters done out of total (for example 6 of 15) | Opens Progress |

| Card | Content | Action |
|---|---|---|
| Store Status | Quick card: active and pending counts. Shows a badge for pending, rejected, expiring or expired payments | Opens Store |
| COC Exam | Green-tinted card: "Test your knowledge across all courses" | Opens the standalone COC Exam module |
| Recently Accessed | Last three items (topic, note or exam) with time ago, and See all | Opens the item after the content lock check |

The three quick cards sit in one row of equal width. Payment alerts have no separate banner: they appear as a badge on Store Status and as a notification. Before any learning, Continue Learning reads "Start your first course" and Recently Accessed shows a hint.

## 5.3 Notes and Amharic Note flow

**Topic/Chapter → Check content lock → Read**

| Condition | Behaviour |
|---|---|
| Active entitlement | Open reader; save position locally |
| Unpaid | Free-trial items open; the rest locked with a Buy action |
| Pending payment | Status shown; locked |
| Expired | Locked; Renew action |
| Unpublished | Unavailable state |

Notes and Amharic Note are two **views of one bilingual content record**. Switching the lower tab keeps the same topic and scroll anchor. Mixed Amharic, English and `$…$` / `$$…$$` formulas render offline.

## 5.4 Practice and Quiz flow

**Topic → Content lock → Practice → Question → Answer → Immediate feedback → Next**

1. Validate entitlement and device.

2. Load question set from cache or server.
3. Student selects an answer; it is written to encrypted SQLite within 10 ms.
4. Correct/Incorrect feedback and explanation appear before Next is enabled.
5. Next never discards the previous answer; Back shows the saved selection.
6. Completion summary shows score and review; attempt is queued in the outbox.

Practice & Quiz 1, Quiz 2 and Quiz 3 have independent question sets, settings and progress.

## 5.5 Mid / Final / Mock flow

**Click → Check content lock → All Questions → Click answer → Save → Navigate → Submit → Result**

1. Lock check; rules screen shows duration and question count.
2. Start creates an attempt and records the monotonic baseline.
3. All Questions screen shows a numbered palette (answered, flagged, unanswered).
4. Each tap saves instantly; the timer runs on elapsed monotonic time.
5. Submit asks for confirmation; auto-submit at zero.
6. Result or status screen (result visibility follows Admin policy).

Exam state machine: `READY → STARTED → IN_PROGRESS → SUBMITTING → SUBMITTED → RESULT_AVAILABLE`, with `IN_PROGRESS → LOCAL_RECOVERY → IN_PROGRESS` on interruption.

## 5.6 COC Exam Flow (standalone module opened from Home)

**Home → COC Exam → COC list (practice sets, exam sets by year) → Check content lock (full or sample) → Practice or Timed Exam → All Questions → Answer → Submit → Result with breakdown by course**

COC Exam is **not** part of the Exams tab and does not belong to one course. It has its own screens, reached only from the Home COC Exam card. Its questions come from one pool that covers all courses, so a set mixes courses. Unpaid students see a sample set and a lock on the rest. Saving, timer, submission and sync follow section 5.5.

## 5.7 Store and Payment flow

**Store → Stream filter → Product → Course selection → Price summary → Payment instructions → Import screenshot → Compress below 300 KB → Transaction reference → Submit → Pending → Result**

| Screen | Content |
|---|---|
| Store home | Stream toggle (Natural / Social / Both), Full Package card, Bundle card, Single courses list, promotional courses |
| Bundle builder | Checklist of semester courses, counter "7–10 selected", live price, extra-course lines |
| Payment instructions | Method tabs (CBE Mobile Banking, CBE Birr, Telebirr), Admin account number with Copy, exact amount |
| Proof upload | Gallery picker, preview, auto-compression to WebP/JPEG under 300 KB, reference field (e.g. `FT…`, Telebirr transaction ID) |
| Status | Pending, Approved, Rejected with reason and Resubmit |

## 5.8 Device lock and release flow

| Scenario | UI |
|---|---|
| First device | Silent bind after login, confirmation toast |
| Second phone | Full-screen modal, no app content reachable: "*ይህ አካውንት በሌላ ስልክ ላይ ተመዝግቧል። ስልክ ለመቀየር እባክዎ ዋናውን ስልክ ያጥፉ/ይውጡ ወይም አድሚኑን ያነጋግሩ*" with buttons Contact Admin and Log Out |
| Release | Original phone: Profile → Security → **Release Device & Log Out** → confirm → binding revoked → local credentials cleared |
| Admin recovery | Admin revokes binding from Devices (reason and audit required) |

## 5.9 Profile Setup, Profile and My Course List

**First run:** Register or first Login → Device check → **Profile page in setup mode** → University → Program → Semester → Stream → Select courses → Save → Home.

**Later edits:** Bottom nav Profile → edit any field or open **My Courses** → Add or Remove → Save → Course tab shows the updated list.

| Profile section | Content |
|---|---|
| Header | Photo, name, phone or email, Edit |
| Academic | University, Program, Semester, Stream (Natural or Social); cascading pickers with search |
| My Courses | Count, list with remove icon, **Add courses** button |
| Subscription and Store | Active products, expiry dates, pending orders, Renew, Buy |
| Security and Device | Bound device name, **Release Device & Log Out** |
| Settings | Language (English, Amharic), downloads and storage, notifications |
| Support | Ticket list, contact Admin |

### Setup rules

1. University, program, semester and stream are required. The Save button stays disabled until all four are set.
2. Program options depend on the university; semester options depend on the program; the course picker depends on semester and stream.
3. The course picker shows courses of the chosen stream plus Common courses, with search, a stream chip filter, a price or lock badge and a selected counter.
4. A student may finish setup with no courses; the Course tab then shows an empty state with **Add courses**.
5. Save writes locally first, queues the change in the outbox and navigates to Home. When editing later, Save returns to Profile with a confirmation and the Course tab updates at once.

### My Course List rules

| Rule | Behaviour |
|---|---|
| Add | Any published course of the student's scope may be added, paid or not. Unpaid courses show a Free badge on their trial portion and a lock on the rest |

| Rule | Behaviour |
|---|---|
| Remove | Confirmation dialog. Progress and answers are kept and return if the course is re-added |
| Paid course removal | Warning that the course is paid; removal hides it but does **not** cancel the entitlement or refund |
| Access | The list never grants access. Only active entitlements unlock content |
| Scope change | Changing university, program, semester or stream asks for confirmation; courses outside the new scope leave the list and their progress is archived |
| Purchase | The bundle builder opens with the courses in the list preselected; the student may adjust to 7–10 |
| Offline | List edits work offline and sync by outbox |

## 5.10 Screen Design Baseline (from approved mockups)

These rules fix the look of the Course screens. Visual tokens are suggested defaults for the design system.

| Element | Specification |
|---|---|
| Look | Light blue-white background, white rounded cards (12–16 dp radius, soft shadow), primary blue for active states, Noto Sans Ethiopic for Amharic text |
| Top app bar | Back arrow, course name (for example English) centred-left, overflow menu (three dots) with Edit course list, Download manager, Help |
| Top tabs | Three pill buttons with icon and label: **Material** (book), **Tests** (clipboard), **Exams** (document check). Active tab is filled blue with white text |
| Lower tabs | Text tabs with a blue underline on the active one; labels are short |
| Bottom navigation | Home, Course, Progress, Profile; Course is active on all course screens |

| Element | Specification |
|---|---|
| List cards | Left icon or logo tile, bold title, one or two lines of grey description, right chevron |

| Screen | Layout and content |
|---|---|
| Material → Modules | Section header "Modules — Study materials and resources"; cards such as English Book (complete textbook and reading materials) and Communication (listening, speaking and communication skills) |
| Material → Notes | Header "Notes — Key concepts and summaries"; cards such as Grammar (rules and examples) and Vocabulary (important words and phrases) |
| Material → Amharic | Header "Amharic Notes — Key concepts in Amharic"; cards such as Translation (English to Amharic). Items are views of the same bilingual records, not separate files |
| Tests → Quiz 1, 2, 3 | Header "Quizzes — Practice and improve your understanding"; chapter cards with a numbered coloured circle, "Chapter N" and the question types (Multiple choice and short answers). Admin assigns chapters to each quiz; the mockup shows Quiz 1 with chapters 1–2, Quiz 2 with 3–4, Quiz 3 with 5–6 |
| Exams → Mid | Header "Mid Examinations — Past midterm exams from universities"; one card per university with its logo, short name and full name (for example AASTU, BDU, INU, Jimma, Mekelle) |
| Exams → Final | Same list under "Final Examinations — Past final exams from universities" |
| Exams → Mock | Same list under "Mock Examinations — Practice with past mock exams" |
| COC Exam (from Home) | Not part of Exams. COC Exam opens from the Home COC Exam card (section 5.6) |

| Screen | Layout and content |
|---|---|
| Locked item | Same card with a lock badge and a Buy or Renew prompt on tap; unpaid courses show Free badges on their trial items |

Tapping a university card opens its papers; tapping a paper runs **Check content lock → All Questions → Answer → Submit** (section 5.5). The student's own university is listed first and a search field filters long lists.

## 5.11 My Courses Screen (search and filter tabs)

The Course tab opens **My Courses**, the student's My Course List with search and filters.

| Element | Specification |
|---|---|
| Top bar | Back arrow, title "My Courses", overflow menu (Add or remove courses, Download manager) |
| Search | Field at the top: "Search your courses…". Filters as the student types by course title or code. Works offline and with Amharic text |
| Filter icon | Right of the search field. Opens a sheet with Stream (Natural, Social, Common) and Access (Paid, Free trial, Locked or expired). A count badge shows active filters; Clear all resets |
| Filter tabs | Pills under the search: **All** · **Ongoing** · **Completed**. The active pill is filled primary blue |
| Course card | Letter tile, course title, subtitle (stream and access state, for example "Common · Paid · 24 days left"), progress bar with percent, chevron. Free-trial and locked badges where they apply |
| Empty states | No matches: message and Clear filters. No courses: message and Add courses |
| Footer action | Add or remove courses (opens the course picker) |

| Tab | Rule |
|---|---|
| All | Every course in My Course List, including courses not started |
| Ongoing | Progress above 0% and below 100% |
| Completed | Progress at 100% |

Search, filter sheet and tab combine: a course must match all three. Progress for a course is chapters completed divided by chapters in the course; a chapter is completed when its notes are read to the end (Admin-configurable rule).

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# 6. Admin Web Portal UI Flow Design (Next.js)

The portal is responsive web, role-protected, English with Amharic content fields. Every sensitive action requires confirmation and writes an audit record.

## 6.1 Payment verification

**Dashboard → Payment Queue → Open receipt → Verify student, product, courses, amount, reference → Inspect image → Approve or Reject (reason mandatory) → Confirm → Entitlements created → Notification → Audit**

The review page shows quoted price against selected courses, duplicate-reference warning, student history and zoomable receipt.

## 6.2 Products, pricing and duration

**Products → Select product → Edit price, course count range, extra-course schedule, duration → Save → Version recorded → Applies to new orders**

Admin manages: bundle price, min/max courses, extra-course price tiers, single-course price, Full Package price, both-streams price, duration per product, promotional flags.

## 6.3 Content publishing

**Course → Module → Topic → Edit bilingual content (English and Amharic fields on one record) → Attach PDF or images → Preview as student → Validate → Publish → New version → Access re-evaluation**

### 6.4 Scan-to-PDF

**Scan/Import → Select 1 to N images (6 or more fully supported) → Reorder by drag → Crop/Rotate/Remove → Preview → Convert to one PDF → Validate → Metadata → Save draft → Review → Publish**

PDF quality gates: page order as set by Admin; correct orientation; no accidental blank pages; page count and file size shown; conversion failure is retryable with the original images retained.

### 6.5 Questions, exams and COC

**Course → Question bank → Create/Import (with LaTeX preview) → Assign to Quiz 1/2/3, Mid, Final, Mock set; COC exams are built separately from the cross-course pool → Set duration, shuffle, feedback → Publish version**

### 6.6 Student, device and entitlement control

Search student → view profile, device binding, entitlements, payment history → suspend, revoke device, grant or extend entitlement (reason required) → audit.

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## 7. Functional Requirements by Partition

Priority: **P0** launch-blocking, **P1** launch-expected, **P2** later. Every requirement is testable and traceable (section 14). Identifiers continue the FR numbering of v1.2 and v1.3; new ranges are FR-120 to FR-159.

### 7.A Partition A — Identity and Device

**Definition:** who the student is and which single phone may use the account. **Workflow:** Register → Login → Read secure device ID → Bind or verify → Session → (Release) → Re-bind.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-001 | P0 | Student registration (name, phone or email, password) | Valid input creates a STUDENT; duplicates rejected |
| FR-002 | P0 | Login with credentials | Valid login issues access and refresh tokens |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-003 | P0 | Session restore | App restart re-checks token and device before showing content |
| FR-004 | P0 | Secure device ID from Android Keystore / iOS Keychain | Every protected request carries `x-device-id` |
| FR-005 | P0 | One-device binding enforced server-side | Second phone gets 403 `DEVICE_MISMATCH_LOCKED`; app is unusable there |
| FR-006 | P0 | **Release Device & Log Out** on the bound phone | Binding revoked, tokens and local credentials cleared |
| FR-007 | P0 | Localized mismatch modal (Amharic and English) | Modal blocks all content; offers Contact Admin and Log Out |
| FR-008 | P0 | Admin inspects and revokes bindings | Reason mandatory; audited |
| FR-009 | P1 | Security event log (login, bind, mismatch, release) | Actor, device hash, result, time stored |
| FR-009a | P0 | Cached content on an unbound or revoked device is inaccessible | Local DB key is derived from the bound device and session; revoked sessions wipe it |

## 7.B Partition B — Academic Catalog and Streams

**Definition:** the University → Program → Semester → Course tree and the stream filter.
**Workflow:** Admin builds catalog → student picks scope → catalog filtered by stream → course card state computed from entitlement.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-010 | P0 | Admin manages universities | Published universities appear in selection |
| FR-011 | P0 | Programs belong to a university | Correct list for the chosen university |
| FR-012 | P0 | Semesters belong to a program | Correct list for the chosen program |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-013 | P0 | Courses belong to a semester | Correct list per semester |
| FR-014 | P0 | Promotional (unpaid) flag per course | Discoverable; protected content locked |
| FR-015 | P0 | Entitlements map to specific courses or packages | Unpaid course never opens as paid |
| FR-016 | P2 | Display ordering | Student lists follow Admin order |
| FR-017 | P0 | Course stream attribute: NATURAL, SOCIAL, COMMON | Every course has exactly one stream |
| FR-018 | P0 | Stream filter in Courses and Store | Natural filter shows Natural plus Common; Social shows Social plus Common |
| FR-019 | P1 | Student may change semester scope | Progress retained per scope; entitlements unaffected |

## 7.C Partition C — Content, Documents and Bilingual STEM

**Definition:** everything a student reads. **Workflow:** Draft → attach assets → preview → validate → publish version → lock check → read → save position.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-020 | P0 | Modules per course | Published modules listed in Modules tab |
| FR-021 | P0 | Topics/chapters per module | Reachable within modules |
| FR-022 | P0 | Notes | Open only through content lock |
| FR-023 | P0 | Amharic Note as a view of the same bilingual record | No duplicate Amharic-only source file exists; tab switch keeps position |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-024 | P0 | Content versioning | Published versions immutable and traceable |
| FR-025 | P0 | Admin scan/import of images | Images listed, editable, reorderable |
| FR-026 | P0 | Image-to-PDF conversion (see Part G) | Valid PDF asset produced |
| FR-027 | P0 | PDF publishing with metadata | Only entitled students download or view |
| FR-028 | P1 | Admin-set free-trial portion per course | Only flagged trial items open; the rest stays locked |
| FR-029 | P0 | **Content lock** check before every protected open | Missing, pending or expired access is locked on server and client |
| FR-034 | P0 | Bilingual STEM rendering: inline `$…$`, block `$$…$$`, mixed Amharic text | Bundled KaTeX fonts and Noto Sans Ethiopic; renders 100% offline with no CDN, clipping or baseline error |
| FR-043 | P1 | Chapter-by-chapter download | No monolithic textbook; memory stays under 120 MB |
| FR-044 | P1 | Reading position saved locally | Continue Learning restores topic and scroll |

## 7.D Partition D — Tests, Exams and the COC Exam

**Definition:** all question-driven activities. **Workflow (tests):** Topic → lock → Practice → Question → Answer → Immediate feedback → Next. **Workflow (exams):** Click → lock → All Questions → Click answer → Submit.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-030 | P0 | Topic practice engine | Topic → lock → practice works |
| FR-031 | P0 | Practice & Quiz 1 | Independent set, settings and tracking |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-032 | P0 | Quiz 2 | Independent |
| FR-033 | P0 | Quiz 3 | Independent |
| FR-035 | P0 | Question engine (single choice, multiple choice, true/false, short answer; LaTeX; images) | Answer persisted on tap |
| FR-036 | P0 | Immediate feedback and explanation | Shown before Next is enabled |
| FR-037 | P0 | Mid exam | Lock checked before start |
| FR-038 | P0 | Final exam | Lock checked before start |
| FR-039 | P0 | Mock exam | Lock checked before start |
| FR-040 | P0 | Zero-loss answers: write to encrypted SQLite within 10 ms | Answer survives kill, crash, airplane mode |
| FR-041 | P0 | Durable attempt lifecycle | Permitted interruption resumes |
| FR-042 | P0 | Idempotent submission | No duplicate final attempt |
| FR-045 | P0 | All Questions palette (answered, flagged, unanswered) | Jump to any question; selections preserved |
| FR-050 | P0 | Anti-cheat timer on monotonic elapsed time (Android `SystemClock.elapsedRealtime()`; iOS equivalent) | Surviving restart; clock rollback cannot extend time |
| FR-051 | P0 | Home COC Exam card | Opens the standalone COC Exam module |
| FR-052 | P0 | Admin manages COC courses and exams | Published COC obeys locks |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-053 | P0 | COC question pool covering all courses, with year, source and version | Provenance visible to Admin |
| FR-054 | P1 | Results stored and shown per Admin policy | Result links to attempt and version |
| FR-055 | P0 | Navigation preserves answers | Back and forward keep selections |
| FR-056 | P0 | Exam lock | Locked exam cannot start |
| FR-057 | P0 | Exam integrity metadata and invalid-state rejection | Attempt audit trail exists |
| FR-058 | P0 | **COC Exam is a standalone module on Home** (not a lower tab of Exams) | Exams shows only Mid, Final and Mock; COC opens only from Home |
| FR-059 | P1 | COC practice mode (untimed, feedback) and exam mode (timed) | Student chooses mode per paper |

Exam timer rules: store `start_elapsed` and `duration`; remaining = duration − (now_elapsed − start_elapsed). After a device reboot the monotonic base resets, so on resume the client sends elapsed-at-pause and the server reconciles against its own start time; the smaller remaining value wins. Detected wall-clock tampering is flagged on the attempt.

## 7.E Partition E — Commerce, Payment and Entitlement

**Definition:** how a student pays and how access is granted, timed and removed. **Workflow:** see section 3.6. Pricing rules PR-1 to PR-7 apply.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-090 | P0 | Store with products, promotional courses and access state | State visible per item |
| FR-091 | P0 | Payment instructions: CBE Mobile Banking, CBE Birr, Telebirr, Admin account, exact amount | Account number is copyable |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-092 | P0 | Screenshot import | Proof enters Pending |
| FR-093 | P0 | Client-side compression to under 300 KB (WebP or JPEG) | Oversize re-processed or rejected with 422 |
| FR-094 | P0 | Transaction reference mandatory ( `FT…` for CBE, Telebirr Txn ID) | Missing reference blocks submit |
| FR-095 | P0 | `UNIQUE(transaction_reference)` in PostgreSQL | Duplicate returns 409 `TRANSACTION_REFERENCE_DUPLICATE` |
| FR-096 | P0 | Admin verification queue | Approve or Reject with reason |
| FR-097 | P0 | Approval activates entitlements for the exact paid scope | Student opens included courses |
| FR-098 | P0 | Admin-managed duration per product | New entitlements use current duration |
| FR-099 | P0 | Automatic expiry lock | Expired access returns 403 `ENTITLEMENT_EXPIRED` |
| FR-100 | P1 | Renewal | Follows the same proof workflow; extends from `expires_at` if still active |
| FR-101 | P1 | Payment and expiry alerts on Home | Alert opens the relevant action |
| FR-102 | P0 | Entitlement lifecycle audit | Actor, time, scope recorded |
| FR-120 | P0 | Bundle builder: select 7–10 courses of one semester and stream for the bundle price | Submit disabled outside range; server revalidates |
| FR-121 | P0 | Admin-managed bundle size range and price | Changes apply to new orders |
| FR-122 | P0 | Extra courses beyond bundle maximum at Admin-managed tiered price (default 100 then 50 ETB) | Price summary itemises extras |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-123 | P0 | Single-course purchase, allowed before any bundle | Entitlement covers that course only |
| FR-124 | P0 | Full Freshman Package per stream (default 350 ETB) | Unlocks all stream and Common courses, all freshman semesters |
| FR-125 | P0 | Buy one stream or both streams in one order | Both-streams price Admin-managed |
| FR-126 | P0 | Stream-filtered Store | Same rules as FR-018 |
| FR-127 | P0 | Server-side price calculation and quote | Client amount is ignored; quote stored on the order |
| FR-128 | P0 | Per-course entitlement rows | Students with course-specific access cannot open unpaid courses |
| FR-129 | P1 | Package supersedes owned single or bundle courses | No double-charge prompt for owned courses |
| FR-130 | P1 | Rejected proof can be corrected and resubmitted | Original rejection reason retained |
| FR-131 | P1 | Renewal window and expiring warnings (default 7 days, Admin-managed) | Alert appears inside the window |
| FR-132 | P0 | Order and receipt history for the student | Statuses and reasons visible |

## 7.F Partition F — Offline and Synchronization

**Definition:** the app keeps working and never loses an answer without internet. **Workflow:** Action → validate locally → SQLite write → UI confirms → outbox row (UUIDv7) → NetInfo online → NestJS sync → acknowledgement → mark complete.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-060 | P1 | Offline cache of permitted content | Cached content usable within the offline grace policy |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-061 | P0 | Offline reading progress | Continue Learning restores position |
| FR-062 | P0 | Outbox of unsynced mutations | Survives restart |
| FR-063 | P0 | UUIDv7 operation IDs | Unique and idempotent |
| FR-064 | P0 | Connectivity worker (NetInfo) | Flushes on reconnect |
| FR-065 | P0 | Idempotent NestJS sync endpoint | Duplicates return already-applied |
| FR-066 | P1 | Deterministic conflict resolution | State converges (matrix in section 11) |
| FR-080 | P0 | Zero answer loss | Local state kept until server acknowledgement |
| FR-081 | P0 | Learning status screen | Progress and completion clear |
| FR-082 | P1 | Honest sync status | UI never claims server sync while pending |
| FR-083 | P0 | Offline entitlement grace | Cached expiry date is honoured; Admin-set grace limit stops indefinite offline use; clock rollback detected with monotonic reference |

## 7.G Partition G — Admin Operations (Next.js Portal)

**Definition:** the single privileged role controlling every asset. **Workflow:** section 6.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-110 | P0 | Dashboard: students, pending payments, content, expiries | Role-protected |
| FR-111 | P0 | Student search, view, suspend, grant or extend access | Audited |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-112 | P0 | Manage universities, programs, semesters, courses, streams | Changes traceable |
| FR-113 | P0 | Manage modules, topics, notes, Amharic content, PDFs | Publishing validates metadata |
| FR-114 | P0 | Manage questions, quizzes, exams, COC (cross-course pool and sets) | Sets versioned |
| FR-115 | P0 | Payment review with duplicate-reference and amount check | Approve or reject audited |
| FR-116 | P0 | Manage products, prices, bundle range, extra tiers, durations, promotions | Rules enforced server-side |
| FR-117 | P0 | Scan-to-PDF workstation | See FR-140 to FR-147 |
| FR-118 | P1 | Send notifications and manage support tickets | Student sees status |
| FR-119 | P0 | Audit log viewer and operational reports | Sensitive actions traceable |
| FR-140 | P0 | Accept 1 to N images (6 or more fully supported; default limit 200) via upload or camera capture | Thumbnails shown |
| FR-141 | P0 | Drag-and-drop **order control** | Final PDF follows Admin order exactly |
| FR-142 | P0 | Crop, rotate, delete, replace page | Edits non-destructive until convert |
| FR-143 | P0 | Convert all images into one PDF | PDF created; page count equals image count |
| FR-144 | P0 | Validate PDF (page count, size, readable, blank-page warning) | Failures are retryable; sources retained |
| FR-145 | P0 | Metadata: university, program, semester, course, category, language, access scope | Required before publish |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-146 | P0 | Draft, review, publish states with versioning | Published PDF inherits entitlement protection |
| FR-147 | P1 | Optional compression and optional OCR text layer | Large scans stay below the Admin size cap |

## 7.H Partition H — Notifications and Support

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-150 | P1 | Receipt submitted, approved, rejected notices | Delivered in-app, push when available |
| FR-151 | P1 | Expiring and expired notices | Sent at configured thresholds |
| FR-152 | P1 | Device mismatch guidance and sync-pending notices | Shown on the correct screen |
| FR-153 | P2 | New content announcements | Admin can target by course |
| FR-154 | P1 | Support request from the app | Ticket visible to Admin |
| FR-155 | P1 | Admin reply to ticket | Student sees reply in Support |

## 7.I Access Matrix

| Feature | Unpaid (free trial) | Paid active | Pending | Expired |
|---|---|---|---|---|
| My Course List | Add any course of the scope | Open | Status | Renew |
| Material (Modules, Notes, Amharic Note) | Free-flagged items open; rest locked | Read | Free items only | Free items only |

| Feature | Unpaid (free trial) | Paid active | Pending | Expired |
|---|---|---|---|---|
| Tests (Quiz 1–3) | Free-flagged chapters | Practise | Free items only | Free items only |
| Exams (Mid, Final, Mock) | Locked unless a sample paper is flagged | Start | Sample only | Sample only |
| COC Exam (Home) | Sample set | Full pool | Sample set | Sample set |
| Store | Buy | Renew if due | Await review | Renew |
| Learning status | Trial progress | Full | Payment state | History plus expiry |

## 7.J Partition I — Profile, Profile Setup and My Course List

**Definition:** the student's academic scope and chosen course list, which drive every other screen. **Workflow:** Register or first login → Profile Setup → Home; later Profile → edit scope or courses → Save → Course tab updates.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-160 | P0 | After registration or first login with an incomplete profile, navigate to the Profile page in setup mode | Home cannot be reached until required fields are saved; then Home opens automatically |
| FR-161 | P0 | Required profile fields: university, program, semester, stream | Save disabled until all are set |
| FR-162 | P0 | Cascading pickers from the published catalog | Program list follows university; semester list follows program |
| FR-163 | P0 | Course picker filtered by scope and stream, Common always included, with search and price or lock badge | Natural filter shows Natural plus Common |
| FR-164 | P0 | Add courses to My Course List | Added course appears in the Course tab at once |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-165 | P0 | Remove courses with confirmation; paid-course warning | Entitlement and progress are untouched |
| FR-166 | P0 | Course tab shows only My Course List with an empty state and Add courses action | No unrelated course is listed |
| FR-167 | P0 | Save navigates to Home in setup mode and back to Profile in edit mode | Navigation matches mode |
| FR-168 | P1 | Confirm scope change and refresh the course list | Out-of-scope courses leave the list; progress archived |
| FR-169 | P0 | Persist the list on the server (`student_course`) and sync through the outbox | Edits made offline reach the server once |
| FR-170 | P1 | Edit name, phone or email, photo, language | Changes saved and audited |
| FR-171 | P0 | Course screens follow section 5.10 (top tabs, lower tabs, cards, bottom navigation) | Design review against mockups passes |
| FR-172 | P1 | Bundle builder preselects courses from My Course List | Student can adjust to 7–10 before paying |
| FR-173 | P1 | Quiz tab lists Admin-assigned chapters per quiz; Exams tabs list past exams by university with search and own university first | Lists match Admin configuration |
| FR-174 | P1 | Admin views and corrects a student's scope and course list with a reason | Audited |

## 7.K Partition J — COC Exam and Free Trial

**Definition:** the cross-course COC module on Home, and the free portion unpaid students can use. **Workflow:** Home → COC Exam → lock check → practice or timed exam → result; unpaid student → add course → open free items → locked item → Buy.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-180 | P0 | COC Exam is a standalone module opened from the Home COC Exam card | It is absent from the Exams tab and from any single course |
| FR-181 | P0 | One COC question pool across all courses; each question carries course, year, source, version | A COC set can mix questions from many courses |
| FR-182 | P0 | Admin builds COC practice sets and exam sets (by year or mixed) | Sets are versioned and publishable |
| FR-183 | P0 | Practice mode (untimed, feedback) and exam mode (timed, monotonic timer) | Same save, submit and sync rules as FR-040 to FR-042 |
| FR-184 | P0 | COC access rule, Admin-configurable: default full pool for any student with an active paid entitlement; unpaid students get a sample set | Rule changes apply without an app update |
| FR-185 | P1 | COC result shows a per-course breakdown | Weak courses are visible |
| FR-186 | P0 | Unpaid students can add courses to My Course List | Course appears with a Free badge on its trial portion |
| FR-187 | P0 | Admin flags free-trial items per course (module, topic, chapter, quiz chapter, sample exam) | Flags are versioned and audited |
| FR-188 | P0 | Default trial rule: first chapter of each module and each quiz free, exams locked, COC sample on | Applies when Admin sets no flags |
| FR-189 | P0 | Server enforces trial flags in the content lock | A forged client cannot open unflagged items |
| FR-190 | P0 | Locked items inside a course show a lock and a Buy prompt | Tap opens the Store with the course selected |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-191 | P1 | Optional trial expiry in days per course | Expired trial locks flagged items except where Admin keeps them free |
| FR-192 | P1 | Trial progress carries over after purchase | No data loss on activation |
| FR-193 | P1 | Trial items cached offline and removed when Admin withdraws the flag | Cache matches current flags after sync |
| FR-194 | P1 | Admin report of trial usage and conversion to purchase | Report filters by course |

## 7.L Partition K — Home Cards and My Courses Search

**Definition:** the card-based Home and the searchable, filterable course list. **Workflow:** Home → card tap → destination; Course tab → search or filter or tab → course → Course screen.

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-200 | P0 | Home is card-based: header, Continue Learning, three quick cards, COC Exam card, Recently Accessed | Order and content match section 5.2 |
| FR-201 | P0 | Greeting with first name, notification bell with unread dot, avatar opens Profile | Dot clears when notifications are read |
| FR-202 | P0 | Continue Learning card shows course, chapter and progress and resumes the last item | Resume rechecks entitlement and device |
| FR-203 | P0 | Quick cards: My Courses (count), Learning Stats (chapters done of total), Store Status (active and pending) | Each opens its destination |

| ID | P | Requirement | Acceptance |
|---|---|---|---|
| FR-204 | P0 | Store Status carries payment alerts (pending, rejected, expiring, expired) as a badge and notification | No separate alert banner exists |
| FR-205 | P0 | COC Exam card opens the standalone COC module | Same behaviour as FR-180 |
| FR-206 | P1 | Recently Accessed lists the last three items with time ago and See all | Tap runs the content lock check |
| FR-207 | P0 | My Courses has a search field, a filter icon and the tabs All, Ongoing, Completed | Layout matches section 5.11 |
| FR-208 | P0 | Search filters by title or code, case-insensitive, offline, Amharic supported | Results update as the student types |
| FR-209 | P0 | Tab rules: All = every listed course; Ongoing = above 0% and below 100%; Completed = 100% | Counts match progress data |
| FR-210 | P1 | Filter sheet: stream and access state, combined with tab and search, with count badge and Clear all | A course must match tab, search and filters |
| FR-211 | P0 | Course card shows tile, title, stream and access subtitle, progress bar and percent, chevron, Free or Locked badge | Badge matches entitlement state |
| FR-212 | P1 | Empty states for no matches and no courses | Each offers the right action |
| FR-213 | P0 | Brand colors and type tokens are unchanged; new cards reuse existing tokens | No new palette is introduced |
| FR-214 | P1 | Course progress = chapters completed divided by total chapters, using an Admin-configurable completion rule (default: notes read to the end) | Percent updates after a chapter completes and syncs |

# 8. Non-Functional Requirements

| ID | Area | Requirement | Acceptance |
|---|---|---|---|
| NFR-001 | App size | Release package under 50 MB | CI reports size and fails the build at 50 MB or more |
| NFR-002 | Memory | Under 120 MB runtime on 2–3 GB RAM phones (Tecno, Infinix class) | Profiled on reference devices |
| NFR-003 | Local write | Under 10 ms for answer persistence | p95 measured on budget device |
| NFR-004 | Offline | Cached content and local answers survive connectivity loss | Airplane-mode test suite |
| NFR-005 | Security | TLS, secure token storage, RBAC, device lock, audit | Security review passes |
| NFR-006 | Localization | English and Amharic across all critical flows | No untranslated critical string |
| NFR-007 | Math | Offline bilingual LaTeX | No network request for fonts |
| NFR-008 | Accessibility | Readable type, 48 dp tap targets, labels | Checklist passes |
| NFR-009 | Scalability | Large assets served from object storage, never bundled | Bundle audit |
| NFR-010 | Observability | Errors, sync lag, payments, locks and conversion failures measurable | Dashboards and alerts exist |
| NFR-011 | Data cost | Receipt under 300 KB; chapter downloads resumable | Verified on throttled network |
| NFR-012 | Availability | Admin and API 99.5% monthly target; daily backups | Restore drill passes |

**Size governance:** Expo production build with Hermes, per-ABI split APKs or Android App Bundle, no bundled textbooks, PDFs or large images, only the required KaTeX fonts and Noto Sans Ethiopic weights (subset where licence allows), dependency size review per pull request.

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# 9. Technical Architecture

## 9.1 System overview

| Layer | Technology | Responsibility |
|---|---|---|
| Student app | React Native with Expo, TypeScript, encrypted SQLite, NetInfo | UI, local-first storage, outbox, timer, KaTeX renderer |
| Admin portal | Next.js (App Router), TypeScript | Control plane, scan-to-PDF workstation, payment queue |
| API | NestJS (REST, JWT, validation, rate limiting) | Business rules, entitlements, sync, audit |
| Database | PostgreSQL 16+, UTF-8 | Authoritative relational state |
| Object storage | S3-compatible bucket | Receipts, PDFs, chapter assets, signed URLs |
| Cache and queue | Redis (local install) | Rate limits, job queue for PDF and notifications |
| Push | Expo push service | Notifications |

The server is the authority for device binding, entitlement, price and final exam acceptance. The client is authoritative only for pending local answers until acknowledged.

## 9.2 Mobile modules (Expo)

`app` (boot, navigation) · `auth` · `device` · `academic` · `home` · `courses` · `material` (reader, KaTeX) · `tests` · `exams` (timer, palette) · coc (standalone COC Exam) · `store` (bundle builder, receipt) · `profile` · `local` (SQLite, cache, outbox) · `remote` (API client, headers) · `shared` (design system, i18n)

### 9.3 Backend modules (NestJS)

`AuthModule` · `DeviceModule` · `AcademicModule` · `ContentModule` · `DocumentModule` (image-to-PDF jobs) · `AssessmentModule` · `CocModule` · `CatalogPricingModule` · `OrderModule` · `PaymentModule` · `EntitlementModule` (expiry scheduler) · `SyncModule` · `NotificationModule` · `SupportModule` · `AdminModule` · `AuditModule`

### 9.4 PDF conversion service

Admin uploads images through signed URLs. A conversion job reads them in Admin order, normalises orientation and size, writes one PDF, computes checksum and page count, stores the result and marks the job `SUCCEEDED` or `FAILED`. Source images are kept until the PDF is published.

### 9.5 Expiry enforcement

Every protected request checks `entitlement.expires_at` against server time. A scheduled job also flips status to `EXPIRED` and queues notifications. Signed asset URLs expire within minutes so a lapsed student cannot keep downloading.

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## 10. Data Model and Integrity (PostgreSQL)

| Entity | Key columns and constraints |
|---|---|
| student | id, phone/email UNIQUE, password_hash, status, stream_pref, university_id, program_id, semester_id |
| device_binding | student_id **UNIQUE**, device_id_hash, status (ACTIVE, RELEASED, REVOKED), bound_at, released_at |
| university, program, semester | Foreign-key hierarchy; publication state; sort_order |
| course | semester_id, code, title, **stream ENUM('NATURAL','SOCIAL','COMMON')**, is_promotional, status |
| module, topic | course/module parent, sort_order |

| Entity | Key columns and constraints |
|---|---|
| content | topic_id, version, status, **title_en, title_am, body_en, body_am** (single record, two language fields), published_at |
| document_asset | storage_key, mime, checksum, size_bytes, page_count, source_job_id, status |
| scan_job, scan_page | job_id, page_no (UNIQUE per job), image_key, rotation, crop; job status |
| question, question_set, exam | course_id, category ENUM(QUIZ1, QUIZ2, QUIZ3, MID, FINAL, MOCK), duration_min, version, is_free_trial |
| attempt, attempt_answer | student_id, exam_id, state, elapsed_ms, server_started_at; answer UNIQUE(attempt_id, question_id) |
| product | type ENUM(BUNDLE, SINGLE, EXTRA, FULL_STREAM, FULL_BOTH), price_etb, duration_days, min_courses, max_courses, stream_scope, is_active |
| price_rule | product_id, tier_no, price_etb (extra-course tiers) |
| orders | student_id, product_id, quoted_amount, stream, status, created_at |
| order_item | order_id, course_id, price_component; UNIQUE(order_id, course_id) |
| payment_proof | order_id, method, image_key, **transaction_reference UNIQUE**, status, reviewed_by, reject_reason |
| entitlement | student_id, course_id, order_id, starts_at, expires_at, status; INDEX(student_id, course_id, expires_at) |
| outbox_event | operation_id (UUIDv7, UNIQUE), student_id, type, payload JSONB, applied_at |
| notification, support_ticket | student_id, type, status, timestamps |

| Entity | Key columns and constraints |
|---|---|
| audit_log | actor_id, action, target_type, target_id, metadata JSONB, created_at |

Integrity rules: every money-affecting write runs in a single database transaction (order, proof status, entitlements); `transaction_reference` is normalised (trimmed, case-folded) before the unique check; entitlement rows exist **per course**, so a Full package writes one row per covered course with the same expiry; all timestamps stored in UTC.

**Added in v2.1:** `student_course` (student_id, course_id, added_at, removed_at, UNIQUE(student_id, course_id)) stores My Course List. It is separate from `entitlement`; removal never deletes an entitlement. `student` keeps `university_id`, `program_id`, `semester_id`, `stream` and a `profile_completed_at` timestamp that drives Profile Setup. `exam` gains `university_id` and `kind` (MID, FINAL, MOCK) so exam lists group by university. `quiz_chapter` (quiz_no 1–3, chapter_id) holds the Admin chapter assignment.

**COC and trial data (v2.2):** `coc_question` (question_id, course_id, year, source, version) forms the cross-course pool; `coc_set` and `coc_set_question` build practice and exam sets; `exam.kind` no longer includes COC. Content, topic, quiz-chapter and exam rows gain `is_free_trial` (boolean); `course` gains `trial_expiry_days` (nullable); `coc_set` gains `sample_question_count`.

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# 11. API Contracts, Error Model and Sync Protocol

## 11.1 Headers

Protected requests send `Authorization: Bearer <token>` and `x-device-id: <hardware_uuid>`. Admin portal requests use cookie or bearer auth and never send a device header; the Admin role is resolved from the server.

## 11.2 Error codes

| HTTP | Code | Meaning |
|---|---|---|
| 401 | UNAUTHORIZED | Missing or invalid token |
| 403 | DEVICE_MISMATCH_LOCKED | Account bound to another phone |

| HTTP | Code | Meaning |
|---|---|---|
| 403 | DEVICE_REVOKED | Binding revoked |
| 403 | ENTITLEMENT_REQUIRED | No active access |
| 403 | ENTITLEMENT_EXPIRED | Duration ended |
| 409 | TRANSACTION_REFERENCE_DUPLICATE | Reference already used |
| 409 | SYNC_CONFLICT | Mutation conflict |
| 422 | RECEIPT_TOO_LARGE | Image over 300 KB |
| 422 | BUNDLE_COUNT_INVALID | Selected courses outside min and max |
| 422 | PDF_CONVERSION_FAILED | Conversion failed |
| 429 | RATE_LIMITED | Too many requests |
| 503 | SERVER_UNAVAILABLE | Retryable |

## 11.3 Key endpoints (catalog)

`POST /auth/register` · `POST /auth/login` · `POST /device/bind` · `POST /device/release` · `GET /universities` · `GET /programs?universityId=` · `GET /semesters?programId=` · `GET /courses?semesterId=&stream=` · `GET /courses/:id/material` · `GET /topics/:id/content` · `GET /exams?courseId=&category=` · `GET /coc/sets` · `POST /orders/quote` · `POST /orders` · `POST /orders/:id/proof` · `GET /entitlements` · `POST /sync/batch` · Admin: `/admin/payments`, `/admin/products`, `/admin/scan-jobs`, `/admin/devices`, `/admin/audit`.

**Profile endpoints (v2.1):** `GET /me/profile` · `PUT /me/profile` · `GET /me/courses` · `POST /me/courses` · `DELETE /me/courses/:courseId` · `GET /courses/picker?semesterId=&stream=&q=` · `GET /exams?courseId=&kind=&universityId=` · `GET /quizzes?courseId=&quizNo=`

## 11.4 Sync protocol

The client sends batches of operations `{operation_id (UUIDv7), type, payload, client_seq}`. The server: (1) validates token and device; (2) confirms the device still owns the account; (3) deduplicates by `operation_id`; (4) validates entitlement, exam

ownership and policy; (5) persists and audits; (6) returns `accepted`, `already_applied`, `rejected`, `conflicted` or `superseded` per operation. The client deletes an outbox row only after acknowledgement.

### 11.5 Conflict matrix

| Data | Rule |
|---|---|
| Practice answer | Latest client timestamp wins within an unsubmitted attempt |
| Exam answer | Accepted only before server-side deadline; late answers rejected and logged |
| Reading progress | Furthest position wins |
| Final exam submit | First accepted submission wins; later ones return `already_applied` |
| Entitlement | Server only; client changes are ignored |

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## 12. Security, Privacy and Content Protection

- TLS everywhere; tokens in secure storage; short access tokens with refresh rotation.
- Server-side checks for device, entitlement, role and price; the client is never trusted.
- Device ID derived from Keystore or Keychain, stored hashed on the server; rooted-device and emulator signals logged for Admin review.
- Encrypted SQLite (key bound to the device and session); paid content stored in app-private storage.
- Signed, short-lived URLs for PDFs, receipts and chapters; Admin-only access to receipts.
- Rate limits on login, registration, payment and sync routes.
- Sensitive actions audited: approvals, price and duration changes, device revocation, publishing, grants.
- Receipts treated as personal financial data: restricted to Admin, retained per policy, never in logs.
- Screenshot and screen-recording blocking on protected readers and exams where the platform allows (Android `FLAG_SECURE`).

- Backups encrypted; restore tested quarterly.

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# 13. Quality Assurance and Release Gates

| Gate | Pass condition |
|---|---|
| Flow | Install → Splash → Login → Device → Profile Setup → Home runs end to end; Profile edit adds and removes courses |
| Device | Second phone returns 403 and shows localized modal; Release then re-bind works |
| Tabs | Course tabs Material, Tests, Exams; Exams has only Mid, Final, Mock; COC Exam opens from Home and draws on all courses |
| Reader | Notes and Amharic Note share one record; formulas render offline |
| Quiz | Immediate feedback; answers persist across kill |
| Exam | Rollback of system clock does not extend time; restart restores attempt |
| Commerce | Bundle 6 and 11 courses rejected, 7 and 10 accepted; extras priced; Full 350; stream filter correct |
| Payment | Receipt under 300 KB; duplicate reference returns 409; approve and reject paths |
| Expiry | Entitlement expires, content locks without app update; renewal restores it |
| Admin PDF | 6 or more images produce one PDF in the Admin-chosen order |
| Offline | Outbox idempotent after repeated flush |
| Size and memory | Package under 50 MB; peak memory under 120 MB on 2 GB device |
| Out of scope | No GPA screen or API exists |

Test pyramid: unit (pricing, entitlement, timer maths), API integration with PostgreSQL, mobile component tests, end-to-end on real budget Android phones, security tests (device bypass, forged role, price tampering), and load tests for payment queue and sync.

**Free-trial and COC gates:** an unpaid student can add a course, open only flagged items, and gets a lock plus Buy prompt on the rest; a forged request for an unflagged item returns 403; COC Exam opens only from Home, mixes courses, and shows a sample set to unpaid students.

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## 14. Traceability and Definition of Done

| Partition | IDs | Mandatory acceptance |
|---|---|---|
| A Identity and device | FR-001 to 009a | Auth, one-device lock, release |
| B Catalog and streams | FR-010 to 019 | Scope selection, stream filter |
| C Content | FR-020 to 029, 034, 043, 044 | Reader, bilingual record, lock |
| D Assessment | FR-030 to 059 | Quizzes, exams, COC, timer, free trial |
| E Commerce | FR-090 to 102, 120 to 132 | Pricing, proof, entitlement, expiry |
| F Offline | FR-060 to 066, 080 to 083 | Zero loss, sync |
| G Admin and scan | FR-110 to 119, 140 to 147 | Control plane, PDF |
| H Notifications | FR-150 to 155 | Alerts and support |

A requirement is done when it is implemented, has automated or manual tests, has server-side authorization, has happy-path and failure UI documented, has database constraints where money or access is involved, has defined offline behaviour, writes audit records for sensitive actions, and ships inside the 50 MB package.

Partition I, Profile and My Course List (FR-160 to FR-174), requires: Profile Setup before Home, course add and remove without touching entitlements, and screens matching section 5.10.

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# 15. Risks, Edge Cases and Mitigations

| Case | Required behaviour |
|---|---|
| Student buys 7 bundle courses then wants an 8th | Allowed up to max 10 inside the bundle; beyond that, Extra Course pricing |
| Student owns a single course then buys Full package | Package covers it; no second charge; expiry extends per package rule |
| Unpaid course opened | Free-trial items open; the rest locked |
| Entitlement expires during exam | Attempt may finish; no new exam can start (Admin policy flag) |
| Phone lost or replaced | Release from old phone, or Admin revokes after identity check |
| Clock changed | Monotonic timer governs exams; server time governs expiry |
| Receipt reused or shared | Unique reference rejects duplicate; Admin sees prior use |
| Fake or edited screenshot | Admin checks amount, account and reference against the bank record; rejection reason retained |
| Wrong amount paid | Admin rejects with reason or approves the partial scope (Admin-configured) |
| Admin approves wrongly | Admin can revoke entitlement with reason; audited |
| PDF conversion failure | Retryable; sources retained |
| Very large PDF | Stored externally; on-demand chapter download |
| App killed mid-quiz | Answers restored from SQLite |
| Low storage | Warn before downloads; evict least-recent chapters |

# 16. Implementation Roadmap

| Phase | Scope | Exit criteria |
|---|---|---|
| 0 | SRS v2.0 approval, UX prototypes, open questions closed | Approved baseline |
| 1 | Monorepo, Expo shell, Next.js admin shell, NestJS, PostgreSQL migrations, CI with size gate | Pipeline green |
| 2 | Identity and device lock | FR-001 to 009a pass |
| 3 | Academic catalog and streams | FR-010 to 019 pass |
| 4 | Content, reader, KaTeX, scan-to-PDF | FR-020 to 044, 140 to 147 pass |
| 5 | Quizzes, exams, COC, timer, free trial | FR-030 to 059 pass |
| 6 | Commerce, pricing, receipts, entitlements, expiry | FR-090 to 132 pass |
| 7 | Offline, outbox, sync | FR-060 to 083 pass |
| 8 | Admin completion, notifications, support, reports | FR-110 to 155 pass |
| 9 | Hardening: security, memory, size, pilot at one university | All release gates pass |

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# 17. Assumptions and Open Questions for Product Owner

| # | Item | Assumption used in this SRS |
|---|---|---|
| Q1 | Does the Full Freshman Package (350 ETB) cover both semesters of the stream? | Yes, because two semester bundles would cost 400 ETB |
| Q2 | Price of a single course | Admin-defined; no default set |
| Q3 | Are extra-course prices 100 for the first and 50 for later ones, or two separate options? | Tiered schedule, Admin may set one flat price |
| Q4 | Price when buying both streams | 2 × 350 unless Admin sets a discount |

| # | Item | Assumption used in this SRS |
|---|---|---|
| Q5 | Do single-course payments count toward a later bundle? | No, unless `credit_single_to_bundle` is on |
| Q6 | Does "all university material" mean every university's version of a paid course? | Yes for the paid course; other courses stay locked |
| Q7 | Can the same phone serve two accounts? | Not specified; this SRS allows it unless Admin enables a unique-device rule |
| Q8 | Is the bundle limited to one semester? | Yes; each semester needs its own bundle |
| Q9 | Default duration per product | Admin-set; suggested one semester (about 150 days) for bundles and one academic year for Full |
| Q10 | Image count for scan-to-PDF | Read as "one or more, six or more fully supported" |

**Added in v2.1**

- **Q11 COC access:** COC Exam is independent of Exams and opens from Home. Default: any student with an active paid course gets the full pool and unpaid students get a sample. Say if full COC needs the Full package or a separate price.
- **Q12 Setup minimum:** the student may finish Profile Setup with no courses selected. Say if at least one course should be required.
- **Q13 Course list size:** no limit on My Course List; the 7–10 rule applies only when buying a bundle.
- **Q14 Exam lists:** exam cards group by university as in the mockup; confirm whether students should see other universities' papers or only their own.

**Added in v2.3**

- **Q15 Not started courses:** they appear only under All. Say if they should also appear under Ongoing.
- **Q16 Progress rule:** a chapter counts as done when its notes are read to the end. Say if quiz completion should also be required.
- **Q17 Payment alerts:** shown on the Store Status card and in notifications, with no banner. Say if you want a banner for expired access.

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# 18. Appendices

## Appendix A — Front-end flow map

`Install → Splash → Login/Register → Device Registration → Profile Setup (university, program, semester, stream, courses) → Home`

Home → Continue Learning · My Courses → Course → **Material** (Modules, Notes, Amharic Note → Topic/Chapter → Lock → Read) · **Tests** (Practice & Quiz 1, Quiz 2, Quiz 3 → Topic → Lock → Practice → Question → Answer → Feedback → Next) · **Exams** (Mid, Final, Mock → Click → Lock → All Questions → Answer → Submit) · COC Exam (standalone, all courses) · Learning Stats · Store Status → Store · Recently Accessed.

## Appendix B — Payment and expiry lifecycle

`Quote → Pay → Screenshot → Compress → Reference → Pending → (Reject → Reason → Resubmit) or (Approve → Entitlements → Active → Expiring → Expired → Locked → Renew)`

## Appendix C — Scan-to-PDF lifecycle

`Images → Scan job → Order and edit → Convert → Validate → Draft → Review → Published → Archived`

## Appendix D — Glossary

| Term | Definition |
|---|---|
| Entitlement | Server-granted access to one course for a period |
| Bundle | 7–10 courses of one semester and stream for one price |
| Extra Course | Course added beyond the bundle maximum at an Admin-set price |
| Full Freshman Package | All courses of a stream for the freshman year |
| Stream | NATURAL, SOCIAL or COMMON classification of a course |
| Promotional course | Unpaid course shown as advertisement |

| Term | Definition |
|---|---|
| Bound device | The single phone allowed to use an account |
| COC Exam | Cross-course COC questions, opened from Home and separate from Exams |
| Content lock | Server and client check that blocks unauthorised access |
| Outbox | Local queue of unsynchronised mutations |
| UUIDv7 | Time-sortable unique operation ID |
| Monotonic clock | Elapsed-time source unaffected by wall-clock changes |
| Scan-to-PDF | Admin conversion of ordered images to one PDF |

## Appendix E — Final acceptance checklist

- Package under 50 MB and memory under 120 MB on a 2 GB phone
- Full front-end flow, Home sections, Course, Material, Tests and Exams tabs exactly as specified
- COC Exam is independent of Exams, opens from Home and covers all courses
- One-device lock and release work; the app is unusable on any other phone
- Stream filter, bundle 7–10 rule, extras, single course, 350 ETB Full package all Admin-configurable
- Manual receipt flow with unique reference and Admin approval
- Duration expiry locks content automatically
- Unpaid courses can be added to the course list and open only their Admin-flagged free-trial portion
- Admin portal in Next.js covers every role asset
- 6 or more images convert to one PDF in Admin order
- Zero answer loss, monotonic timer, offline KaTeX
- No GPA feature

**Added in v2.1**

- After registration or first login the student completes Profile Setup (university, program, semester, stream, courses) and then lands on Home

- Profile lets the student change scope and add or remove courses; the Course tab shows only that list
- Course, Material, Tests and Exams screens match section 5.10 (bottom navigation Home, Course, Progress, Profile)

**Added in v2.3**

- Database is PostgreSQL and the repository uses npm workspaces
- Home is card-based (section 5.2); My Courses has search and the tabs All, Ongoing, Completed (section 5.11)
- Brand colors and fonts are unchanged